

Automated DNS Sinkholing using Wazuh Active Response, DNS-Stats, and AlienVault OTX

Project Overview

In modern SOC environments, DNS is one of the earliest and most reliable indicators of compromise. Malware, phishing campaigns, and command-and-control (C2) infrastructures almost always rely on DNS resolution before establishing outbound communication.

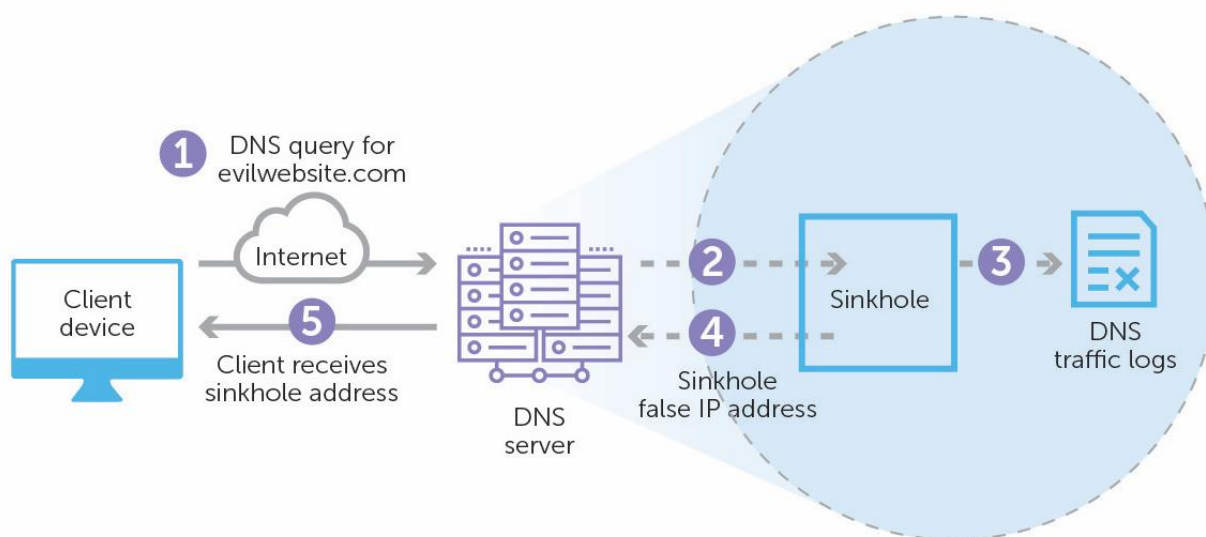
This project is an **extension of my previous DNS threat-hunting pipeline** built using **Wazuh, DNS-Stats, and AlienVault OTX**.

In the previous project, the workflow was **detection-only**, where analysts were alerted when a Windows endpoint resolved a malicious or suspicious domain.

In this enhanced project, I implemented **automated response and containment** using **DNS sinkholing**. Now, when a malicious domain is detected via AlienVault OTX, the endpoint is **immediately protected** by automatically redirecting the malicious domain to a safe internal address — preventing outbound communication with attacker infrastructure.

This transforms the SOC workflow from **alert-only monitoring** to **real-time automated threat response**.

What is a DNS sinkhole?



What is DNS Sinkholing?

DNS sinkholing is a defensive technique where malicious domains are deliberately resolved to a controlled, non-malicious IP address instead of their real destination.

Instead of blocking an IP address (which attackers can easily change), sinkholing **blocks the domain itself**, forcing attackers to register new infrastructure.

In this project:

- Malicious domains are automatically resolved to **127.0.0.1 (localhost)**

- Outbound traffic is silently neutralized
- The endpoint never reaches the attacker

This technique is widely used in **enterprise SOC**s, **IR teams**, and **malware containment strategies**.

Tools & Technologies Used

- Wazuh SIEM
- Wazuh Windows Agent
- Sysmon (Event ID 22 – DNS Queries)
- DNS-Stats (Domain reputation & frequency analysis)
- AlienVault OTX (Threat Intelligence)
- PowerShell
- Wazuh Active Response
- MITRE ATT&CK Framework

Architecture Overview

Detection → Enrichment → Validation → Automated Response

1. Windows endpoint generates DNS activity
2. Sysmon logs DNS queries (Event ID 22)
3. Wazuh ingests and correlates DNS telemetry
4. DNS-Stats identifies rare / first-seen domains
5. AlienVault OTX validates malicious indicators
6. Wazuh Active Response triggers DNS sinkholing
7. Endpoint is automatically protected

Step-by-Step Implementation (Mapped to Screenshots)

Extending the Previous DNS Detection Pipeline

This project builds directly on the previously completed **DNS-Stats + AlienVault OTX integration**. All detection, enrichment, and alerting workflows were already functional.

The focus of this project is **adding automated response**, not rebuilding detection from scratch.

Goal:

Move from “*alerting analysts*” → “*automatically protecting endpoints*”.

Step 1 – Creating the DNS Sinkhole PowerShell Script (Endpoint)

A PowerShell script was created on the Windows endpoint to:

- Receive alert data from Wazuh Active Response

- Extract the malicious domain detected by AlienVault OTX
- Automatically append the domain to the Windows hosts file
- Redirect the domain to 127.0.0.1

This ensures the malicious domain can no longer resolve to its real IP.

Purpose:

Implement endpoint-level DNS sinkholing.

```
malicious_domains.ps1
File Edit View

#####
##Script to resolve malicious domain to localhost
#####
# Read the Alert that triggered the Active Response in manager and convert to Array
$INPUT_JSON = Read-Host
$INPUT_ARRAY = $INPUT_JSON | ConvertFrom-Json
$INPUT_ARRAY = $INPUT_ARRAY | ConvertFrom-Json
$ErrorActionPreference = "SilentlyContinue"

#Extract Domain base_indicator.indicator
$malicious_domain = $INPUT_ARRAY."parameters"."alert"."data"."base_indicator"."indicator"

#Resolve Malicious Domain to Localhost
Add-Content -Path $env:windir\System32\drivers\etc\hosts -Value "`n127.0.0.1`t$malicious_domain" -Force
```

Step 2 – Creating the Active Response Command Wrapper

A .cmd wrapper script was placed in the Wazuh agent's Active Response directory.

This wrapper:

- Executes the PowerShell sinkhole script
- Allows Wazuh to run PowerShell actions securely under SYSTEM context

Purpose:

Enable Wazuh Active Response to execute endpoint remediation actions.

```
domains.cmd
File Edit View

:: Simple script to run malicious domain PShell script.
:: The script executes a powershell script and appends output (resolve malicious ip to 127.0.0.1).
@ECHO OFF
ECHO.

"C:\Program Files\PowerShell\7\pwsh.exe -executionpolicy Bypass -File "C:\Users\Abdul Rehman\Downloads\malicious_domain.ps1"

:Exit
```

Step 3 – Configuring Active Response on the Wazuh Manager

An Active Response block was added to the Wazuh manager configuration (ossec.conf).

The response is configured to trigger **only when AlienVault confirms a malicious domain**.

- Rule ID: **100080**
- Trigger source: AlienVault OTX

- Execution location: Local endpoint

Purpose:

Ensure automated response is triggered only for confirmed malicious IOCs.

```
<command>
  <name>malicious_domains</name>
  <executable>domains.cmd</executable>
  <timeout_allowed>no</timeout_allowed>
</command>

<active-response>
  <disabled>no</disabled>
  <rules_id>100080</rules_id>
  <command>malicious_domains</command>
  <location>local</location>
</active-response>
```

Triggering a Malicious DNS Request from the Endpoint

After completing the DNS sinkhole configuration, a known malicious domain from **AlienVault OTX** was selected for testing.

The domain was queried directly from the Windows endpoint using PowerShell:

ping acmetoy.com

At this point:

- The DNS query is generated on the endpoint
- Sysmon immediately logs the DNS request (Event ID 22)
- The endpoint attempts to resolve the domain for the **first time**

This single action initiates the **entire detection + response workflow automatically**.

Purpose:

Trigger DNS telemetry and validate real-time SOC automation.

```
PS C:\Users> ping acmetoy.com
```

```
Pinging acmetoy.com [204.16.169.54] with 32 bytes of data:
Reply from 204.16.169.54: bytes=32 time=290ms TTL=50
Reply from 204.16.169.54: bytes=32 time=267ms TTL=50
Reply from 204.16.169.54: bytes=32 time=245ms TTL=50
Reply from 204.16.169.54: bytes=32 time=246ms TTL=50
```

```
Ping statistics for 204.16.169.54:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 245ms, Maximum = 290ms, Average = 262ms
```

Step 6 – Initial DNS Resolution Observed in Sysmon (Pre-Response Visibility)

In the **Sysmon Event Viewer**, Event ID **22 (DNS Query)** confirms:

- The queried domain: acmetoy.com
- The original external IP address returned by DNS
- The process responsible (PING.EXE)
- The user and endpoint context

This shows that the endpoint **attempted to resolve a malicious domain**, which is the earliest possible indicator of compromise.

Important:

This visibility exists **only for milliseconds** before the automated response takes effect.

Purpose:

Provide forensic visibility into attempted malicious communication.

The screenshot displays the Sysmon Event Viewer interface. At the top, a table lists several events, with Event ID 22 (DNS Query) selected. Below this, the 'Event 22, Sysmon' details are shown. The 'General' tab is active, displaying the following information:

Dns query:
RuleName: -
UtcTime: 2026-02-05 09:58:06.731
ProcessGuid: {00a0ddf8-69ae-6984-dc54-000000001d01}
ProcessId: 19508
QueryName: acmetoy.com
QueryStatus: 0
QueryResults: ::ffff:204.16.169.54;
Image: C:\Windows\System32\PING.EXE
User: LAPTOP-20EH9HNF\Abdul Rahman

Below the details, the 'Log Name' is Microsoft-Windows-Sysmon/Operational. The 'Source' is Sysmon, 'Logged' is 05-02-2026 15:28:08, 'Event ID' is 22, 'Task Category' is Dns query (rule: DnsQuery), 'Level' is Information, 'Keywords' are, 'User' is SYSTEM, 'Computer' is LAPTOP-20EH9HNF, and 'OpCode' is Info.

Step 7 – Alert Correlation in Wazuh (Detection Chain)

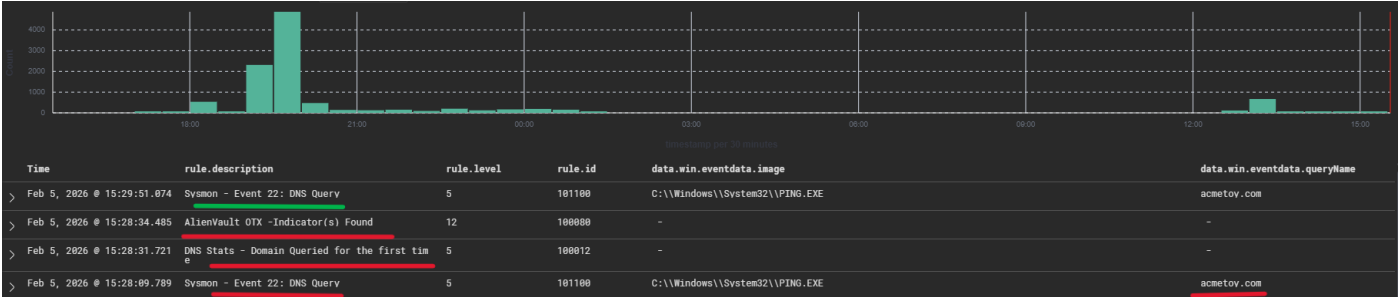
In the Wazuh dashboard, alerts appear **in a strict sequence**, confirming proper correlation:

1. **Sysmon – DNS Query Detected**
 - Confirms endpoint-level DNS activity
2. **DNS-Stats – Domain Queried for the First Time**
 - Identifies the domain as rare / suspicious
3. **AlienVault OTX – Indicator(s) Found**
 - Confirms the domain is malicious based on threat intelligence

This validates the **multi-layer detection pipeline** before any response is executed.

Purpose:

Ensure detection accuracy before triggering automated action.



t	data.win.eventdata.queryName	acmetoy.com
t	data.win.eventdata.queryResults	::ffff:204.16.169.54;
t	data.win.eventdata.queryStatus	0
t	data.win.eventdata.user	LAPTOP-20EH9HNF\Abdul Rehman
t	data.win.eventdata.utcTime	2026-02-05 09:58:06.731
t	data.win.system.channel	Microsoft-Windows-Sysmon/Operational
t	data.win.system.computer	LAPTOP-20EH9HNF
t	data.win.system.eventID	22
t	data.win.system.eventRecordID	599287
t	data.win.system.keywords	0x8000000000000000
t	data.win.system.level	4
t	data.win.system.message	"Dns query: RuleName: - UtcTime: 2026-02-05 09:58:06.731 ProcessGuid: {00a0ddf8-69ae-6984-dc54-000000001d01} ProcessId: 19508 QueryName: acmetoy.com QueryStatus: 0 QueryResults: ::ffff:204.16.169.54; Image: C:\Windows\System32\IPING.EXE User: LAPTOP-20EH9HNF\Abdul Rehman"

t	data.dnsstat.query	<u>acmetoy.com</u>
t	data.dnsstat.seen_by_isc	RDAP
t	data.dnsstat.seen_by_web	Thu, 08 Apr 2004 18:47:54 GMT
t	data.dnsstat.seen_by_you	Thu, 05 Feb 2026 09:58:31 GMT
t	data.integration	dnsstat
t	decoder.name	json
t	id	1770285511.210166068
t	input.type	log
t	location	dns_stats
t	manager.name	wazuhserver
t	rule.description	DNS Stats - <u>Domain Queried for the first time</u>
#	rule.firedtimes	1
t	rule.groups	dnsstat, dnsstat_alert
t	rule.id	100012
#	rule.level	5
🕒	rule.mail	false
t	rule.mitre.id	T1071
t	rule.mitre.tactic	<u>Command and Control</u>
t	rule.mitre.technique	<u>Application Layer Protocol</u>

t	data.base_indicator.indicator	acmetoy.com
t	data.base_indicator.type	domain
t	data.dns_requested	acmetoy.com
t	data.message	Alienvault OTX API response successfully parsed
t	data.sections	general, geo, url_list, passive_dns, malware, whois, http_scans
t	data.status	success
t	data.type	domain
t	decoder.name	json
t	id	1770285514.210196557
t	input.type	log
t	location	alienvault
t	manager.name	wazuhserver
t	rule.description	AlienVault OTX -Indicator(s) Found
#	rule.firedtimes	2
t	rule.groups	alienvault, otx_ioc
t	rule.id	100080
#	rule.level	12

Automated DNS Sinkhole Triggered (No Manual Action)

Immediately after AlienVault OTX confirms the domain as malicious:

- Wazuh Active Response is triggered
- The PowerShell sinkhole script runs on the endpoint
- The malicious domain is automatically added to the Windows hosts file
- The domain is redirected to the configured sinkhole IP (127.0.0.1)

This happens **automatically**, without analyst or user intervention.

Purpose:

Contain the threat at the endpoint in real time.

Step 9 – Re-Testing the Same Domain After Sinkholing

The same domain is queried again from the endpoint:

ping acmetoy.com

This time:

- The domain no longer resolves to the malicious external IP
- It resolves directly to **127.0.0.1**
- The endpoint cannot reach the attacker infrastructure

This confirms that the DNS sinkhole is **actively enforcing protection**.

Purpose:

Verify successful automated containment.

```
PS C:\Users> ping acmetoy.com

Pinging acmetoy.com [127.0.0.1] with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Step 10 – Sysmon Validation After Sinkholing

Sysmon Event ID 22 now shows:

- The same domain (acmetoy.com)
- DNS resolution returning **127.0.0.1**
- No outbound connection to the malicious IP

This proves that DNS resolution has been **permanently altered at the endpoint level**.

Purpose:

Confirm sinkholing effectiveness using endpoint telemetry.

Level	Date and Time	Source	Event ID	Task Category
Information	05-02-2026 15:29:49	Sysmon	22	Dns query (rule: DnsQu...
Information	05-02-2026 15:29:26	Sysmon	22	Dns query (rule: DnsQu...
Information	05-02-2026 15:28:34	Sysmon	22	Dns query (rule: DnsQu...

Event 22, Sysmon

General

Details

Dns query:
RuleName: -
UtcTime: 2026-02-05 09:59:47.916
ProcessGuid: {00a0ddf8-6a13-6984-fb54-00000001d01}
ProcessId: 21168
QueryName: acmetoy.com
QueryStatus: 0
QueryResults: ::ffff:127.0.0.1;
Image: C:\Windows\System32\PING.EXE
User: LAPTOP-2OE9H9HNF\Abdul Rehman

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon Logged: 05-02-2026 15:29:49
Event ID: 22 Task Category: Dns query (rule: DnsQuery)
Level: Information Keywords:
User: SYSTEM Computer: LAPTOP-2OE9H9HNF
OpCode: Info

Step 11 – Wazuh Dashboard Verification

In Wazuh alert details and tables:

- The malicious domain is clearly visible
- AlienVault OTX confirms IOC match
- MITRE ATT&CK mapping shows **Command and Control**
- No further alerts show communication with the malicious IP

Analysts can clearly see:

- What domain was queried
- Why it was malicious
- That it was successfully contained

Purpose:

Provide SOC analysts with full visibility and confidence in response actions.

t	data.win.eventdata.queryName	acmetoy.com
t	data.win.eventdata.queryResults	::ffff:127.0.0.1;
t	data.win.eventdata.queryStatus	0
t	data.win.eventdata.user	LAPTOP-20EH9HNF\Abdul Rehman
t	data.win.eventdata.utcTime	2026-02-05 09:59:47.916
t	data.win.system.channel	Microsoft-Windows-Sysmon/Operational
t	data.win.system.computer	LAPTOP-20EH9HNF
t	data.win.system.eventID	22
t	data.win.system.eventRecordID	599346
t	data.win.system.keywords	0x8000000000000000
t	data.win.system.level	4
t	data.win.system.message	"Dns query: RuleName: - UtcTime: 2026-02-05 09:59:47.916 ProcessGuid: {00a0ddf8-6a13-6984-fb54-000000001d01} ProcessId: 21168 QueryName: acmetoy.com QueryStatus: 0 QueryResults: ::ffff:127.0.0.1; Image: C:\Windows\System32\PING.EXE User: LAPTOP-20EH9HNF\Abdul Rehman"

Step 12 – Final Confirmation in Windows HOSTS File

The Windows hosts file confirms the automated action:

127.0.0.1 acmetoy.com

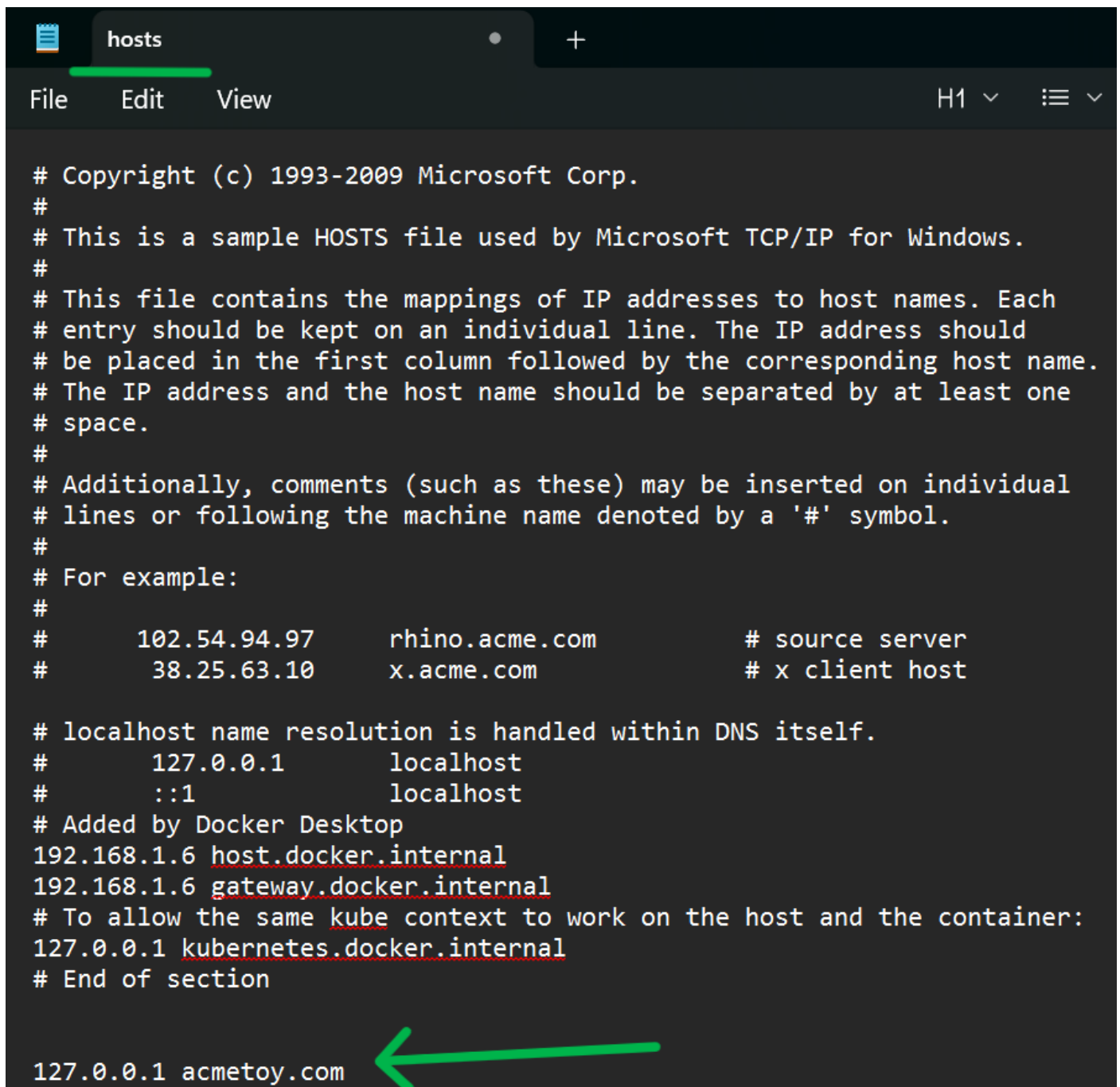
This entry was:

- Added automatically
- Triggered only after AlienVault confirmation
- Enforced locally on the endpoint

This ensures **persistent protection** against the malicious domain.

Purpose:

Validate endpoint-level DNS sinkholing implementation.



```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97       rhino.acme.com          # source server
#       38.25.63.10       x.acme.com              # x client host

# localhost name resolution is handled within DNS itself.
#       127.0.0.1        localhost
#       ::1              localhost
# Added by Docker Desktop
192.168.1.6 host.docker.internal
192.168.1.6 gateway.docker.internal
# To allow the same kube context to work on the host and the container:
127.0.0.1 kubernetes.docker.internal
# End of section

127.0.0.1 acmetoy.com
```

Project Outcome

- ✓ Real-time detection of malicious DNS activity
- ✓ Automated endpoint containment
- ✓ No manual analyst intervention required
- ✓ Domain-level blocking instead of IP-based blocking
- ✓ SOC-style automation using Active Response

SOC & Analyst Value

- Prevents command-and-control communication

- Reduces Mean Time to Respond (MTTR)
- Eliminates reliance on perimeter firewalls
- Provides deep visibility into DNS behavior
- Enables proactive threat containment

Skills & Concepts Demonstrated

- SIEM engineering (Wazuh)
- DNS-based threat hunting
- Threat intelligence integration
- Active Response automation
- Endpoint security controls
- PowerShell for security automation
- SOC alert correlation & tuning
- MITRE ATT&CK mapping
- Detection → Response lifecycle design

Conclusion

This project demonstrates a **real-world SOC automation workflow** where DNS telemetry is not only detected and enriched, but **acted upon automatically**.

By extending my previous DNS-Stats and AlienVault integration with **DNS sinkholing**, I transformed a detection-only pipeline into a **fully automated response system** — closely mirroring how modern enterprise SOCs detect, validate, and contain threats in real time.
